

Measuring AI Ability to Complete Long Software

50% task-completion time horizon doubling every ~7 months

Kwa, West, Becker, et al. — METR | arXiv:2503.14499 | March 2025

AI Safety Researchers · Frontier Lab Leadership ·
Policy Analysts

A New Metric: 50% Task-Completion Time Horizon

An intuitive, human-grounded measure of real-world AI autonomous capability

01

Assemble Task Suite

Curate 170 diverse software & research tasks spanning 3 seconds to 8 hours of skilled human effort.

02

Measure Humans & AI

Time skilled human baselines (>800 humans, 2,529 hours) and measure AI agent success rates.

03

Fit Logistic Model

Estimate the task duration at which each AI agent achieves 50% success — the time horizon.

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?" Choices: "run.sh", "run.txt", "run.py", "run.md"
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

Key Insight

This metric enables direct comparison across vastly different models — from GPT-2 to Claude 3.7 Sonnet — using a single human-grounded scale.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

Three benchmark families combined with ~800 human baselines (2,529 hours)

97 **HCAST**

Diverse software tasks

1 min – 30 hrs

7 **RE-Bench**

Difficult ML research engineering

~8 hrs each

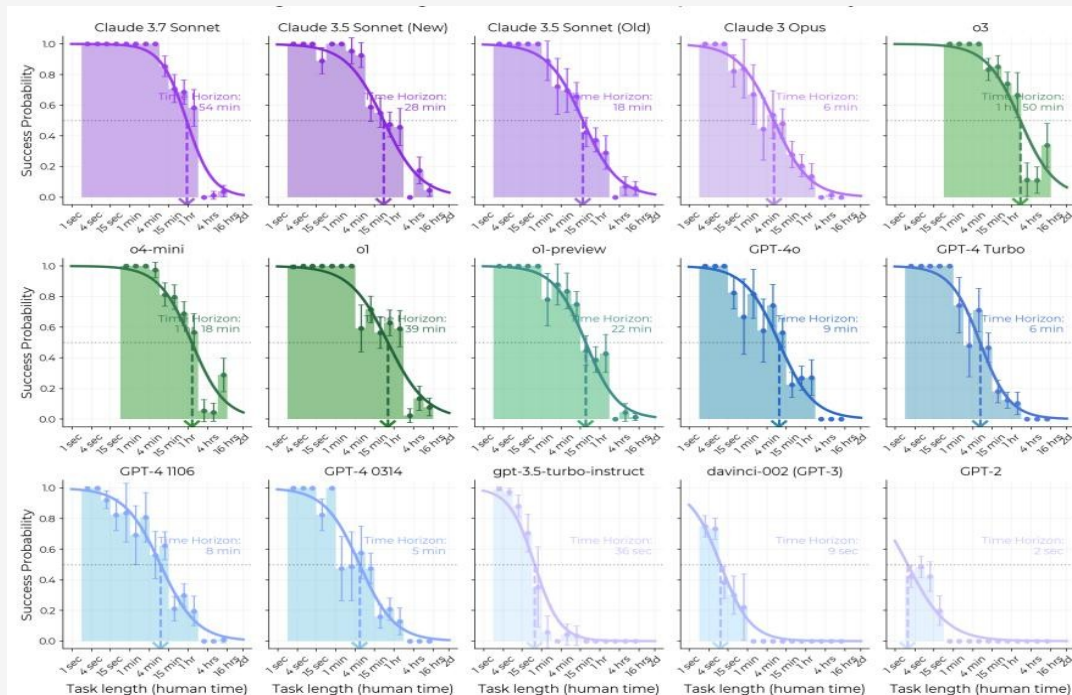
66 **SWAA**

Short single-step actions

1 sec – 30 sec

Task Family	Human Time	Description
find_shell_script	3 seconds	Multiple choice: Which file is a shell script?
wikipedia_research	1 minute	Research simple factual information from Wikipedia
oxdna_simple	9 minutes	Detect and fix a bug in molecular dynamics simulation input files
munge_data	56 minutes	Write a Python script to transform JSON data by inferring conversion rules
cuda_backtesting	8 hours	Implement custom CUDA kernels for 30x speedup of a backtesting tool

AI Task Horizon Grew from 2 Seconds (GPT-2) to ~1 Hour 50 Minutes (o3) in 6 Years



Source: METR, arXiv:2503.14499

~207 days
Doubling time (~7 months)

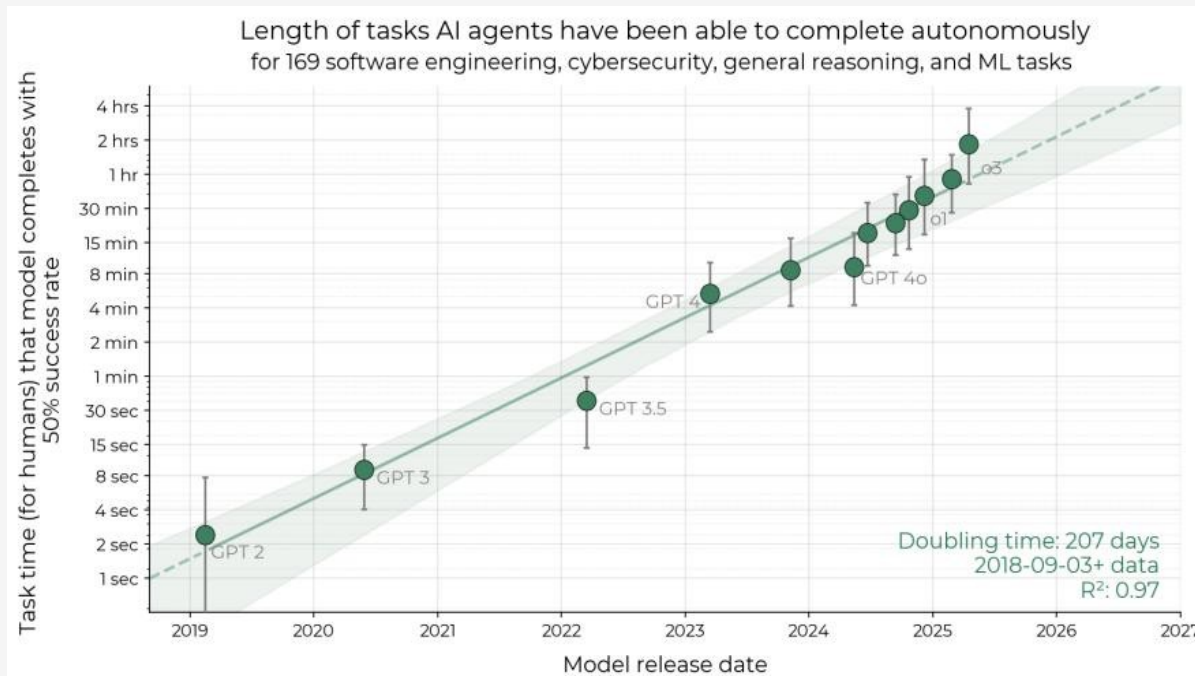
$R^2 = 0.97$
Exponential fit over 6 years of
model releases

Key Milestones

- GPT-2 (2019): ~2 seconds
- GPT-4 (2023): ~5 minutes
- o3 (2025): ~1 hour 50 minutes

Success Probability Drops Sharply as Task Length Increases for All Models

Per-model logistic curves show frontier models pushing the 50% threshold rightward



Source: METR, arXiv:2503.14499

50% Time Horizon by Model

o3	1h 50m
Claude 3.7 Sonnet	54 min
o1	39 min
Claude 3.5 Sonnet	28 min
GPT-4o	9 min
GPT-4 (0314)	5 min
GPT-3.5-turbo	36 sec

The 80% time horizon follows a similar trend but is roughly 5x shorter than the 50% horizon.

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

What enables models to handle longer, more complex tasks autonomously?



Improved Logical Reasoning

Better chain-of-thought and structured problem decomposition let models plan multi-step solutions across longer horizons.



Better Tool Use

More reliable invocation of code interpreters, terminals, and APIs extends effective capability without raw parameter growth.



Reliability & Self-Awareness



Adaptation to Mistakes

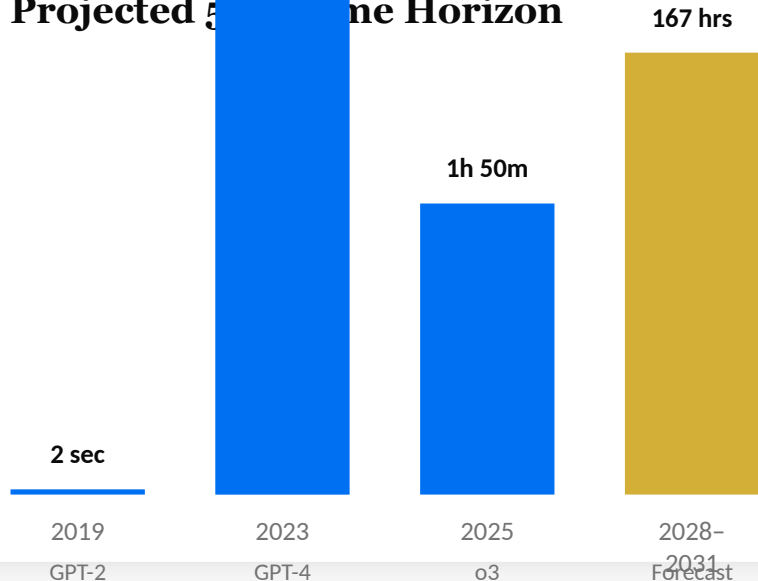
Limitation

Performance is notably lower on less structured, “messier” tasks. External validity remains an open question — the benchmark tasks may not perfectly represent the average segment of real-world intellectual labor.

Extrapolation Predicts Month-Long Task Automation by 2028–2031

If the exponential trend continues, frontier AI could autonomously complete month-long software projects

Projected Software Project Time Horizon



Forecast Assumptions & Caveats

- Naive exponential extrapolation of the 2019–2025 trend predicts a 1-month (167 work-hour) time horizon between mid-2028 and mid-2031.
- The 2023–2025 growth rate is ~20% faster than the overall 6-year rate, suggesting possible acceleration.
- External validity is uncertain: benchmark tasks may not generalize to all real-world job distributions.

Implication

Month-long software project automation would represent a qualitative shift in autonomous AI capabilities and labor substitution.

Takeaways: Exponential Autonomous Capability Growth Demands Proactive Safety Governance

01

Robust Metric

The 50% time horizon is a robust, intuitive metric for tracking AI autonomy, with $R^2 = 0.97$ over 6 years of model releases.

02

Doubling Every ~7 Months

The time horizon doubles roughly every 207 days, with possible acceleration since 2024.

03

Frontier at ~2 Hours

Current frontier models (o3) autonomously complete tasks that take skilled humans nearly 2 hours — up from 2 seconds for GPT-2.

04

External Validity Open Question

Benchmark tasks may not perfectly represent real-world intellectual labor; supplementary experiments found little evidence of slower trends on realistic tasks.

05

Governance Imperative

Key limitations include poor performance on unstructured tasks and uncertain generalization. Proactive safety governance is essential as month-long automation approaches.

Source: Kwa, West, Becker, et al.

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